Lab 10: Piezoelectric Accelerometer

- Review
  - Vibration of a cantilever
    - Exact method
    - Crude approximation for natural frequency
    - Other approximate methods
      - Rayleigh-Ritz method
      - Assumed modes
      - Finite element
      - etc
  - Piezoelectric accelerometer
    - Equivalent circuit (see spec sheet)
- Equipment
  - Digital Multimeter Pro DMM (link)
  - National Instruments USB 6341(link)
  - Breadboard with triple power supply
  - Sweep/function generator (link)
  - Oscilloscope (link)
  - 741 op amp (link)
    - $V_S^+ = 6$ V
    - $V_S^- = -6$ V
  - OP07 op amp (link)
    - $V_S^+ = 6$ V
    - $V_S^- = -6$ V
  - INA121PA-ND op amp (link)
    - $V_S^+ = 6$ V
    - $V_S^- = -6$ V
  - Resistors:
  - Accelerometer - Metrix Instrument Co.
    - **Please do not directly connect a corroded BNC connector to oscilloscope**

- In our Instrumentation and Controls lab we have several piezoelectric accelerometers. The accelerometers are old and show signs of corrosion. **The objectives of this lab are to: (i) determine experimentally the bending stiffness of your assigned cantilever beam (EI) and (ii) determine if the manufacturers calibration (voltage sensitivity) of your assigned accelerometer remains accurate.**

- Lab report:
  - Your lab report should follow the required lab report structure for this class.
  - In the body of your report, you should build a strong case for your objectives. **The objectives of this lab are to:**
    - **(i) determine experimentally the bending stiffness of your assigned cantilever beam (EI)**
      - Quantify the uncertainty/error in the quantities you obtain
      - Justify your experimental procedure: circuit, sample rate, filter, number of samples, etc.
      - Outline the mathematical theory used in your analysis
      - How does your measurement of stiffness compare to the expected stiffness for the given geometry and material?
    - **(ii) determine if the manufacturers calibration (voltage sensitivity) of your assigned accelerometer remains accurate**
      - What is the manufacturers stated voltage sensitivity?
      - What is the voltage sensitivity you measure?
      - Quantify the uncertainty/error in the quantities you obtain
      - How do these sensitivities compare? Are they within measurement uncertainty?
      - Justify your setup and procedure: circuit, sample rate, filter, number of samples/trials, etc.
      - Clearly state your conclusion
  - Include the following in the Appendix:
    - This lab outline (required)
    - Data you collected during lab (required)
    - Screenshot of each LabView block diagram and/or Matlab scripts used to acquire/process data (required)
    - Responses to the questions throughout the lab outline. (optional)
    - Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

References